

ORIGINAL ARTICLE

Classification Algorithm for Skin Color (CASCo): A new tool to measure skin color in social science research

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Abstract

Objective: A growing body of literature reveals that skin color has significant effects on people's income, health, education, and employment. However, the ways in which skin color has been measured in empirical research have been criticized for being inaccurate, if not subjective and biased.

Objective: Introduce an objective, automatic, accessible and customizable Classification Algorithm for Skin Color (CASCo).

Methods: We review the methods traditionally used to measure skin color (verbal scales, visual aids or color palettes, photo elicitation, spectrometers and image-based algorithms), noting their shortcomings. We highlight the need for a different tool to measure skin color

Results: We present CASCo, a (social researcher-friendly) Python library that uses face detection, skin segmentation and k-means clustering algorithms to determine the skin tone category of portraits.

Conclusion: After assessing the merits and shortcomings of all the methods available, we argue CASCo is well equipped to overcome most challenges and objections posed against its alternatives. While acknowledging its limitations, we contend that CASCo should complement researchers' toolkit in this area.

KEYWORDS

colorism, measurement, photo elicitation, racism, skin color, spectrometers

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A recent review of academic literature demonstrates that there is a globalized preference for white skin, attributable to racism associated with European colonization (Dixon and Telles 2017). A growing body of empirical research reveals that skin color has significant effects on people's income, health, education, and employment (Arce, Murguía, and Frisbie 1987; Hughes and Hertel 1990; Keith and Herring 1991; Monk 2015; Weaver 2012). There are, however, two shortcomings in this literature. The first one is that, while variations in human skin color have been recorded for centuries, most of the empirical research on its social impacts has been limited to the United States (Dixon and Telles 2017; Jablonski 2012; Loveman M 2014; Malhotra and Raso 2007). The second issue has to do with the way in which skin color has been measured and recorded. The measurement methods employed thus far have been criticized for being inaccurate, if not subjective and biased (Hannon and DeFina 2014; Hersch 2008; Hill 2002a, 2002b). We think our contribution has the potential to redress these two shortcomings, at least partially.

In this article, we introduce CASCo—Classification Algorithm for Skin Color—, a new tool to measure skin color. CASCo is objective, automatic, accessible, and customizable. On the one hand, having an accessible and customizable tool facilitates the replication and democratization of this kind of research: it can be adapted to fit the parameters required by researchers in diverse contexts. On the other, CASCo constitutes an objective and automated technique to measure skin color and therefore overcomes the challenges that personal biases pose to researchers. We do not claim CASCo will solve all the difficulties of conducting empirical research on the social impacts of skin color; there are, of course, limitations to this tool. However, CASCo represents a valuable tool to measure skin color, which should be available for social researchers.

To achieve its purposes, the article is structured as follows. In the section titled “Existing Approaches to Measuring Skin Color”, we conduct a review of the methods that (social) scientists have used to measure skin color in the past, outlining their merits and shortcomings. These methods include verbal scales, color palettes, photo elicitation, spectrometers, and image-based algorithms. Appendix 1 summarizes our review of methods and related works. In the section “Classification Algorithm for Skin Color (CASCo)”, we introduce the technical details of this new tool. We also present an example to demonstrate its ease of use and how to interpret its results. In the section: “Assessing CASCo”, we discuss CASCo's benefits and limitations. We conclude that CASCo represents a significant contribution to feasible, replicable, and objective measurement of skin color. The merits of this method, however, need to be properly understood in order to avoid errors and farfetched interpretations.

EXISTING APPROACHES TO MEASURING SKIN COLOR

Verbal scales

This method refers to studies and surveys that ask respondents (or interviewers) to classify a subject into categories or verbal scales that represent a color spectrum, usually ranging from “dark” to “light” or “black” to “white.” This method is probably the most used in social research, both in the United States and beyond. Surveys using this method to measure skin color include (but are not limited to) the U.S. Census (1850–1920), the National Chicano Survey (1979), and the National Survey of Black Americans (1979).

Widespread as it is, this approach goes not without objection. On the one hand, some have criticized how these verbal cues skew and “determine” the answers they seek to find. Harris et al. (1993) added “morena” to a split census and found out that the proportion of respondents who self-identify as being of mixed-color race increased dramatically, while those identifying as white or black substantially declined. They argue that their findings uncover the error of imposing rigid categories deemed appropriate for establishing racial identity (Harris et al. 1993). On the other hand, the studies using this method to measure skin color have overlooked the fact that interviewers are not neutral and detached recorders of the respondents' physical traits. Statistical analyses have revealed a compelling “race-of-the-interviewer effect.” Hill (2002a) demonstrates that both black and white Americans exhibit relatively limited ability to judiciously differentiate the physical features of people from other races.

Self-reported measures face similar issues; they may not accurately reflect how respondents are seen by others and may be, instead, the result of a “reverse causality” through which experiences—including perceived discrimination—determine the perception of one’s own color (Monk 2015). Furthermore, studies have shown that self-identification often differs from observers’ classification (Campbell and Troyer 2007; Perreira and Telles 2014) and that racial self-identifications are subject to change (Golash-Boza and Darity 2008, p. 904).

Color palettes

In response to the limitations of verbal scales, several surveys have provided interviewers with visual aids or color palettes to use as benchmarks for different skin tones. These color palettes are, arguably, exogenous indicators for skin color; and many have utilized them as an alternative to self-perception and interviewer-based measurements (Perreira and Telles 2014, pp. 241–242). Surveys employing this method include the New Immigrant Survey, the 2012 General Social Survey, the AmericasBarometer, and the Project on Ethnicity and Race in Latin America (PERLA).

Using visual aids or color palettes as benchmarks might be helpful for the interviewer, especially if they receive training to mitigate the impact of racial bias in their categorization. However, this method is still susceptible to other factors that may distort one’s perception of skin color. For example, in the United States, men in professional attire are more likely to be categorized as “white,” while people of low socioeconomic levels are more likely to be categorized as “black” (Garcia and Abascal 2016, p. 423). This is what Harris (1964) calls “money whitening”: a process by which affluent individuals are more likely to self-identify—and be classified—in a “whiter” category. This hypothesis has been tested and confirmed in Latin America too, where studies show that one’s racial status is determined, at least in part, by one’s social status (Golash-Boza and Darity 2008, p. 904).

Photo elicitation

Photo elicitation is a qualitative method in which photographs are used to generate verbal discussion (Thomas 2009). Researchers typically present image(s) to participants and then ask for comments. This produces a different kind of insight as it evokes feelings and memories, not only information (Harper 2002). To measure skin color using this method, respondents are asked to classify photos according to their skin color or to identify the color closest to their own (Dixon and Telles 2017).

Given its qualitative nature, this method has been used primarily in studies with small samples, but some large exercises have employed it too, such as the 2002 Brazilian Social Survey. Photo elicitation has been used to reveal and understand emic categories based on skin color (Gravlee 2005; Roth 2012), racialized notions of beauty (Rondilla and Spickard 2007; Sorokowski, Sorokowska, and Kras 2013), and racial profiling (Aguilar 2011; Candelario 2007; Eberhardt et al. 2004).

Despite its clear merits, the method has been criticized on the grounds that other elements contained or associated to the photos affect how people perceive skin color. For instance, Garcia and Abascal (2016) demonstrated that skin color ratings were affected by the presence of a racially distinctive name. Other potential biases include underestimation of the variation when evaluating outgroup members, using of oneself as a reference category, pressures related to what is social “desirable,” and the influence of other physical traits and contextual factors (Hill 2002b).

Spectrometers and spectrophotometers

Although slightly different in technical specifications, spectrometers and spectrophotometers are similar enough to be grouped together for the purpose of this review. To perform the measurement, the light of a

lamp “passes through glass color filter of known optical property, and through an aperture in the photocell, and then impinges on the surface to be measured; the light which is diffusely reflected from the surface acts on the photocell” (Lasker 1954, p. 112). For accurate measurements, the areas where measurements are taken should be free of pigmented hairs, clean, and accessible and present a flat surface to the search unit—usually the forehead and the medial intermuscular septum in the middle of the upper arm (Lasker 1954, pp. 115–116). When employed correctly, these tools achieve objective, reliable, and standardized measurements of skin color, without any human bias.

However, the number of projects using spectrometers is relatively small and mostly limited to medical research. We think three main issues make this technology inaccessible and unfeasible for social research. First, spectrometers tend to be expensive equipment. Even if their price has decreased thanks to technological advances, it still represents a significant investment, which restricts the number of research projects able to utilize them. Second, spectrometers are complex technological devices; it takes interdisciplinary collaboration to operate them, process the measurements, and explain the results. Third, even if one has the material and human resources to use the equipment, there are external and uncontrollable factors that can distort the results. For instance, Lasker (1954) reported that fluctuations in the local power and deviations in the frequency of the alternating current affected his measurements. He explicitly stated that this technique is “too cumbersome for use in field situations” (Lasker 1954, pp. 115–116). The practicalities of research make it difficult, if not impossible, to use this method in different settings and on a large scale.

Image-based algorithms

The appearance of human faces changes constantly due to internal (i.e., emotions, facial hair) and external factors (i.e., orientation, lightning, scale). These changes challenged face recognition technology until very recently. However, computerized face detection has advanced dramatically, boosted by the potential profits from applications such as videoconferencing, surveillance, identity authentication, and human–machine interaction.

Currently, there are several face detection algorithms (see Appendix 1 for some examples) that use AdaBoost, pixel classification, and connectivity analysis. Interestingly, one of the cues scientists used to perfect the technique was skin color, as it provides valuable information to distinguish a face from other elements of the picture (Mahmoud 2008, p. 501). There are many color spaces (i.e., RGB; LUV; LAB; XYZ; HIS, HSV), which an algorithm could use to detect skin color in an image, but studies have shown that RGB is not suitable for accurately measuring skin color (Wu and Ai 2008). Contrarily, HSV gives the best performance for skin color detection (Sandeep and Rajagopalan 2002; Zarit, Super, and Quek 1999), and it has been used to design face and self-adaptive face detection algorithms (Liu and Peng 2010).

Even if rarely, image-based algorithms have already been used in social research. These tools certainly provide objective measurements and escape the objections posed to other methods. However, we find three issues with the way social science has incorporated the use of this technology: It has chosen ineffective algorithms, it has been secretive, and it has not been user-friendly. Most of the studies available fall in (at least) one of these categories.

For instance, Robst et al. (2011) examine the relationship between skin color and salary in professional basketball players. To measure skin tone, they use an RGB colorscore, derived from a model in which red, green, and blue are combined in various ways to reproduce other colors. While we applaud the innovation to use an objective measure of skin tone, studies have found that the RGB color space is not suitable for constructing accurate skin color models because it fails to clearly distinguish between mere color and the intensity of a pixel (Chandrappa, Ravishankar, and Rameshbabu 2011; Wu and Ai 2008, p. 339). Furthermore, the authors do not share the code for their algorithm, which makes their study hard to evaluate and replicate. Other studies that claim to use algorithms as part of their methodology to measure skin color do not share the codes either (Campos-Vazquez and Rivas-Herrera 2021).

In sum, the technology to detect and objectively measure skin color accurately exists, but the opportunities it represents have not been fully seized within social science. The algorithms used are mostly

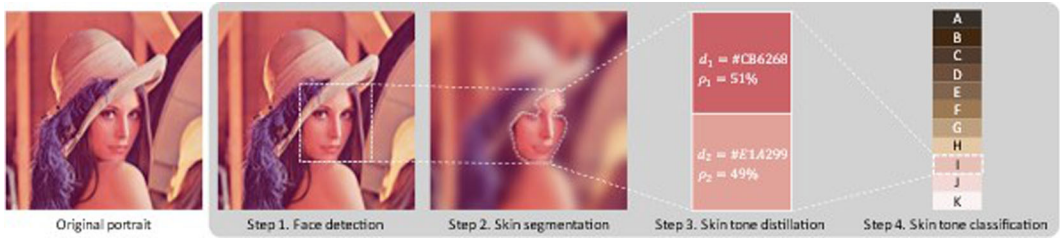


FIGURE 1 Classification Algorithm for Skin Color's (CASCo) color detection and classification process.

restricted; this is not only because the software is not published or expensive to buy, but because it continues to be specialized and highly technical. This makes it inaccessible for most social scientists and, even more, activists.

In the next section, we unpack CASCo, an objective, automatic, accessible, and customizable algorithm to measure skin color.

Classification Algorithm for Skin Color (CASCo)

In response to the shortcomings and limitations of the skin color measurement methods reviewed in this article, we introduce the CASCo.

Specifications

This Python library is open source, and it can be used to detect face and skin colors in predetermined areas of specified images. The process is objective, automatic, and customizable. To determine the skin tone category of specific portraits, CASCo works based on face detection, skin segmentation, and k-means clustering algorithms.

In Figure 1, we use the Lenna picture (Hutchinson 2001) to demonstrate how our model:

1. Detects the face area: As portraits can contain different body parts, to get consistent classification results, CASCo focuses on the skin tone of the face. First, it maps the face area color space into the HSV space, which is more in line with the way human vision perceives color properties (Chen, Chen, and Chien 2008).
2. Extracts non-skin areas: Given a specified threshold in HSV color space, non-skin areas (i.e., eyes, hair, and teeth) are filtered out from the face using a Gaussian blur function (Jones and Rehg 2002).
3. Distills dominant colors: CASCo employs the k-means clustering algorithm to distil $|D| = 2$ dominant colors in the remaining skin area.
4. Assigns a category: CASCo assigns the portrait to a customizable category (c^*) with the closest color detected from the portrait based on the minimum weighted Delta E (CIE 2000) distance (ΔE_{00}) (Sharma, Wu, and Dalal 2005):

$$c^* = \arg \min_{c \in C} \sum_{d \in D} \rho_d \cdot \Delta E_{00}(d, c), \quad (1)$$

where C is the set of color categories, $c \in C$; D is the set of detected dominant colors in the specified portrait, $d \in D$; ρ_d is the proportion of color d and $\sum_{d \in D} \rho_d = 1$.

In its default configuration, CASCo will assign the portrait to a category of the PERLA project. PERLA, based at Princeton University, created this color palette with 11 items to facilitate skin tone studies in Latin America, and it has been widely used in social research (Campos-Vazquez and Medina-Cortina 2019; Rejón Piña 2022; Telles, Flores, and Urrea-Giraldo 2015). We preconfigured this palette into CASCo to support future research in a region where the topic continues to be understudied; however, the color palette in which to categorize the portraits is completely customizable as well.

5. Results: After performing its functions, CASCo creates a report file (*results.csv*), which contains nine columns with detailed information. For our illustration:

file	face_location	dominant_1	props_1	dominant_2	props_2	category	PERLA	distance(0-100)
lena_std	84:153	#CB6268	0.51	#E1A299	0.49	#e7c1b8	I	17.37

where **file** is the filename of the processed image; **face_location** are the coordinates (top_left : bottom_left) of the detected face area; **dominant_x** are the HEX values of x dominant color in the detected face area; **props_x** are the proportion of x dominant color in the detected face area; **category** is the HEX value of the assigned color category; **PERLA** is the label of the assigned color category in the PERLA color palette; and **distance(0-100)** is the distance between the assigned color category and the dominant colors in the face area. Values are between 0 and 100. A lower value indicates a more accurate classification.

Running this algorithm would barely require coding or programming literacy, given that a pseudocode of CASCo is available in Appendix 2, and the full code is publicly available. <https://anonymous.4open.science/r/CASCo>

Demonstration

Academic studies have revealed that, in Mexico, skin color has statistical impact on variables as diverse as literacy, income, education level, and social mobility (Arceo-Gomez and Campos-Vazquez 2014; Campos-Vazquez and Medina-Cortina 2019; Narro and Moctezuma 2012); and some experiments suggest that skin color might have an impact on the success of political candidates (Aguilar 2011). However, the topic of political representation remains severely understudied. Researchers have reported a lack of reliable and standardized data on the matter (Htun 2015; Sue, Riosmena, and Telles 2021; Telles 2004) as one of the obstacles preventing this research. CASCo can contribute to redress this situation by harvesting data in a quick and reliable manner. Using an example, in this section, we demonstrate the ease of use of our tool and how to interpret the results.

Through its Information System for Parliamentary Statistics, the Mexican Chamber of Deputies publishes information of its composition and about the performance of the legislators. As mentioned before, the data available does not include any variables related to race, ethnicity, or skin color. However, since 2006, the Chamber publishes a photo album of the legislators. For this demonstration, we use CASCo to process the [album](#) of the LXIII Legislature (2015–2018).

To extract the portraits from the .pdf file, one can use any image extractor tool (some of which are available online and for free). To make the analysis easier, these images should be assigned a title (ID) and stored in one folder. This ID will be used to match CASCo's results to the relevant case in a database. For this example, the folder "Portraits_LXIII" contains the 500 photos (titled 1 to 500) of the legislators, and it is stored in the "Downloads" folder of the computer.

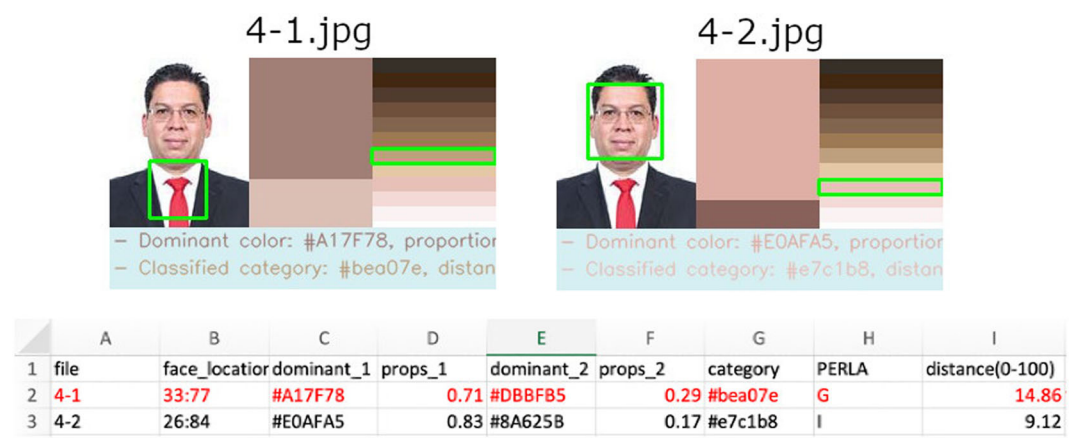


FIGURE 2 CASCo's processed files and *results.csv*.

CASCo is a Python library; therefore, to use it, one needs to access a Python environment. For users with no programming experience, we recommend installing and using Miniconda. Once installed, open the shell/terminal and enter the command `pip install -U skin-tone-classifier`, which will install CASCo (confirm the version of the library is $\geq 0.1.9$). Then, navigate to the folder where the images are stored by entering: `cd ~ / Downloads`.

After that, run CASCo by entering `-i ./Portraits_LXIII -d`. In this command, `-i` is used to specify the location of the images to process; `-d` means debug mode, in which CASCo will generate a folder with processed images for us to verify if the result is correct. Once CASCo is done, the Downloads folder will contain three new items (1) a *debug* folder with all processed images, (2) a *log* folder with the running logs, and (3) a spreadsheet *results.csv* with the summary of each image.

In the *debug* folder, one can find the files of the processed images with a suffix (i.e., `-1` or `-2`). If an image has more than one file, CASCo has detected more than one “face” in the portrait; in these rare cases, manually verify the correct face and delete all incorrect cases from *results.csv*. In Figure 2, we can see that in our example, CASCo detected two “faces” for the photo 4.jpg, so it created the files 4-1.jpg (which fails to identify the face) and 4-2.jpg (which identifies the face correctly). Reflecting this, *results.csv* contains two rows for the same photo, so we need to delete the row for 4-1.jpg, to discard the wrong reading.

In this example, once *results.csv* is clean without these duplicates, there are 500 rows: one for each legislator. With these data alone, using the PERLA column, one can analyze the “racial” composition in the Chamber of Deputies.

For instance, Figure 3 clearly reveals that light-skinned legislators dominated the LXIII Legislature, while people with darker skin tones were dramatically underrepresented. This is just one use of the many analyses that the CASCo-generated skin color variable makes possible. This variable can be easily added to databases, which include more details about legislators, such as gender and political party. More sophisticated analyses could use this variable to examine whether skin color has an impact on legislative performance.

ASSESSING CASCo

In this section, we discuss the benefits and potential shortcomings of CASCo. While we acknowledge CASCo is not without limitations, we contend that its merits outweigh its shortcomings and that CASCo should complement researchers’ toolkit in this area.

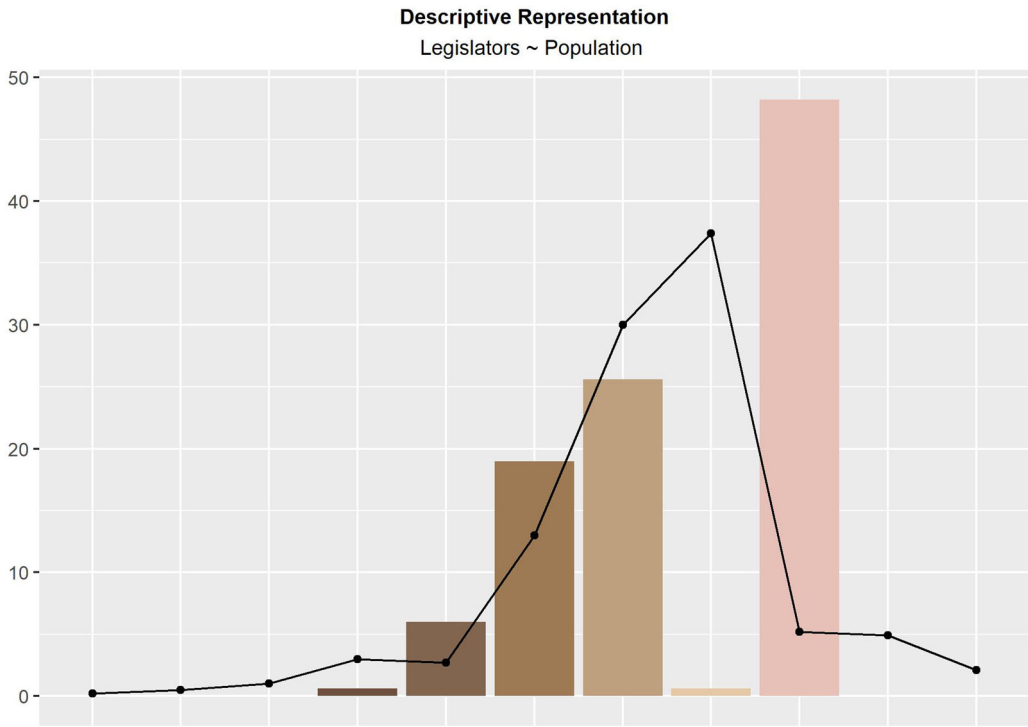


FIGURE 3 LXIII Legislature composition by skin color.

Benefits

- Objective:** As demonstrated in the section “Existing Approaches to Measuring Skin Color”, other methods to measure skin color overlook the interviewers’ and respondents’ own biases. However, CASCo’s measurement and classification into categories does not involve any human assessment. While the parameters are predefined, the measurement and classification processes are completely conducted by the algorithm and based on objective standards, which support accuracy. For instance, it uses the HSV color space (instead of RGB), and the dominant colors are calculated after excluding “distorters” (i.e., hair, eyes, lips) from the image and identifying the colors with the largest proportions in the image. Similarly, the sorting into categories depends not on human appreciation but on the mathematical distance between colors.
- Automatic:** CASCo’s measurement process is completely automatic; this has two main benefits. On the one hand, as reviewed in the Existing Approaches to Measuring Skin Color section, other methods to measure skin color are susceptible to external factors impacting the interviewer’s perception of skin color, but CASCo overcomes these challenges not only because the algorithm has no knowledge or socialization about these external factors, but because it conducts the skin color measurement in complete isolation from other factors and details. In fact, the researcher does not find out about the skin color of a subject until this has already been measured, captured, and stored. Altering or changing the measurement after that point will not constitute bias but a clear ethical offense and a violation to academic rigor. On the other hand, CASCo can quickly process large data sets. Other methods depend on someone conducting the interview, applying a survey, or operating a spectrometer. While some methods are more expensive than others, they are all time-consuming and resource-intensive; but CASCo needs not to conduct the analyses one by one; the measurements can happen simultaneously and efficiently, enabling researchers to engage with much larger samples and data sets.

- *Accessible*: CASCo is not only free, but it has been designed with social scientists in mind. It is easy to use, and it has been preconfigured with parameters that are adequate for a wide range of research projects. Being *free* means that funds and financial resources will not be salient factors for researchers when selecting a technique to measure skin color as it has been the case in the past (see Monk 2015). CASCo is cheaper than its alternatives, without compromising accuracy in measurements. Furthermore, being *social scientist-friendly* is one thing that truly sets CASCo apart from other accurate methods, such as spectrometers and cryptic algorithms. The accessibility of CASCo makes it a valuable tool to conduct replicable and comparable research in different contexts. CASCo has the potential to democratize social research on skin color and to generate valuable data sets for researchers and activists all around the world.
- *Customizable*: Harris et al. (1993) criticized how researchers based in the United States wanted to “export” their conceptual frameworks and racial categories to other settings. CASCo escapes this criticism because all of its variables and parameters are completely and easily customizable. The algorithm requires not to use the same number of dominant colors or even the same categories. CASCo supports any number of categories and based on their distance to a specified color, it will sort portraits accurately, objectively, and automatically. The researchers just need to know the HEX values or the RGB codes of the colors they want to use.

Limitations

- *The makeup objection*: It could be argued that one limitation of this method is that it can be “confused” or distorted by subjects wearing makeup. What is worse, this distortion could disparately affect the skin color measurements in women and could impact the cross-tabulation and correlation analyses of skin color and other variables, such as gender. We concede that CASCo would measure “makeup color” rather than actual skin color, but, if this really constitutes a limitation, this is not exclusive to CASCo. As demonstrated in our review, interviewers are often influenced by external factors, makeup and other class-signifiers included. If anything, CASCo offers a consistent measurement, which is only affected by this external factor, successfully excluding others. Furthermore, the distortion makeup causes could not be as significant for the purposes of social research investigating discrimination anyways because makeup could “whiten” subjects (Rondilla and Spickard 2007) and potentially mitigate the impact of skin color. If subjects usually present themselves in social or professional settings using makeup, then measuring that “makeup color” rather than actual skin color could better explain their experiences of discrimination (of course, another discussion on why subjects want to whiten themselves would still be necessary). Research demonstrates that skin color “as seen by others is probably more important than identity when examining racial inequality and discrimination” (Telles 2012, p. 1167).
- *Photo-based*: One of the main limitations of CASCo is that it requires an image to perform the skin color measurement. This, logically, affects the nature of the research it can assist with and hinders some of its benefits. For example, if a research project does not have access to a data set containing images already, researchers will need to collect (relatively similar) photographs of the subjects they want to study. This might not require sophisticated or expensive equipment (a mobile phone would do), but it does complicate data collection. These challenges are not impossible to overcome, and the associated costs of gathering images might still be cheaper than the other methods; still, we consider this to be a significant limitation to CASCo. In any case, to maximize the benefits of CASCo, researchers could limit their studies to data sets that already contain this kind of visual material. We think this, albeit specific, could still constitute significant research. For example, our demonstration in the CASCo section is a good testament to the fact that even limiting research to preexisting data sets, a lot could be done in terms of analyzing representation, both in popular media and political structures.

CONCLUSION

We have introduced the CASCo, a new tool to measure skin color, which is objective, automatic, accessible, and customizable. We reviewed the methods traditionally used to measure skin color, outlining their merits and shortcomings, to argue CASCo is well equipped to complement researchers' toolkit in this area. First, the accuracy of verbal scales is weakened by the effect of race on either respondents or interviewers. Second, while the use of color palettes and visual aids to classify skin subjects' skin color somehow mitigates the impact of racial bias, respondents are still susceptible to other factors that distort one's perception of skin color, such as clothing, makeup, and jewelry. Third, photo elicitation might be a good tool to prompt and facilitate respondents' answers, but it is affected by potential biases such as the underestimation of the variation when evaluating outgroup members, use of oneself as a reference category, or pressures related to social desirability. Fourth, spectrometers provide an objective measure for skin color, but they are impractical, expensive, and demand specialized training. Last, studies that employ imaged-based algorithms to measure skin color hardly share the codes and usually resort to inefficient algorithms that are also hard to replicate.

CASCo is not the panacea of empirical research on the social impacts of skin color, and the limitations we foresee for this method have been outlined here. However, we sustain that CASCo is a significant contribution to feasible, replicable, and objective measurement of skin color. By virtue of being accessible and customizable, CASCo facilitates the replication and democratization of research. Being an objective and automated method makes CASCo overcome the challenge of personal biases that affect and distort skin color measurements conducted by interviewers or subjects themselves.

While we value the merits of CASCo, we think it is imperative to properly understand that this method measures skin color and nothing else. This distinction might be obvious to some, but it is important to avoid errors and far-fetched deductions. CASCo and its classifications or measurements can be salient, but not definitive, to analyses of racial discrimination. Skin color is just but one of the factors at play in the phenomenon of race; societal values, cultural traits, physical attributes, diet, region of ancestry, institutional power relationships, and education are all components of the aggregate of "race" (Sen and Wasow 2016, p. 506; see also Banton 2012; Wade 2012). This does not mean skin color measurements are irrelevant or purely descriptive because they often elicit a racial ideology (or a "racial schema") where people are aware of human color variation and they often place on a naturalized hierarchy of worth (Roth 2012, p. 12; Telles 2012). Furthermore, skin color is often the most important predictor of racial identification, even above social status (Banton 2012; Telles and Paschel 2014). In fact, in contexts like Latin America, skin color does a better job of capturing social inequality than do racial categories (Telles, Flores, and Urrea-Giraldo 2015). Therefore, sociological studies of skin color measurement should take a multidimensional approach to color and race; we think CASCo constitutes a useful tool for that purpose.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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